Duo Xu

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Education

University of Science and Technology of China Hefei, Anhui 2024.09 – Present

M.Eng. Candidate Computer Technology, expected graduation in June 2027

Harbin Engineering University Harbin, Heilongjiang 2020.09 – 2024.06

B.Eng. Computer Science and Technology

Internship / Project Experience

Tencent Project Up | Hunyuan Image Model Inference Acceleration 2026.01 – 2026.06

- **Challenge**: The post-training stage of Hunyuan Image requires high-throughput and low-latency online inference services to support reinforcement-learning data generation and evaluation. The original inference pipeline was tightly coupled with the training pipeline, leading to limited serviceability, high framework upgrade costs, and complex train-inference consistency validation.
- **Hunyuan Image 3.0 Adaptation**: Migrated the Hunyuan Image 3.0 inference pipeline to vLLM 0.11.0 for reinforcement-learning-oriented training scenarios. Implemented model loading, request construction, generation parameter passing, output parsing, and service interface adaptation, while ensuring numerical and behavioral consistency with the original training pipeline.
- **Hunyuan Image 3.5 Text Generation Optimization**: Adapted the text generation module of Hunyuan Image 3.5 to vLLM 0.19.0. Refactored tokenizer and processor logic, sampling configurations, batched request handling, and service interfaces to support large-scale online text generation during post-training.
- **Hunyuan Image 3.5 Image Generation Optimization**: Adapted the image generation module to vLLM-Omni to support complex multimodal inputs, long invocation chains, and image-related output processing. Implemented image input processing, generation result delivery, and service invocation within the reinforcement-learning training pipeline.
- **Engineering Design**: Adopted a plugin-style and non-intrusive adaptation strategy for vLLM and vLLM-Omni, minimizing modifications to the upstream codebase and reducing maintenance costs for future version upgrades and synchronization with open-source implementations.
- **Outcome**: Completed inference framework adaptation, accuracy alignment verification, and end-to-end service pipeline integration for Hunyuan Image 3.0/3.5, providing reusable high-performance inference services for post-training workloads.

ByteDance | Two-Batch-Overlap vLLM Communication Overlap Optimization 2025.09 – 2025.12

- **Challenge**: In a 2-card NVIDIA L20 environment with PCIe Gen4 interconnect, PyTorch Profiler analysis showed that NCCL AllReduce communication accounted for 50%–55% of total forward-pass time in dense model inference, making it the primary performance bottleneck.
- **Technical Innovation**: Designed a Two-Batch Overlap computation-communication overlap mechanism based on vLLM 0.11.0. The mechanism exploits inter-batch independence in Transformer inference to enable parallel execution, dynamically activates the TBO path for large-batch scenarios, and preserves the original forward path for small batches to avoid unnecessary optimization overhead.
- **Solution**:
  - Designed a batch-splitting strategy for multi-request inference serving, using a token-count-based balancing algorithm to preserve request integrity, achieve over 95% load balance, and maintain memory alignment to multiples of 256.
  - Built an asymmetric dual-stream parallel execution architecture with a three-level synchronization system across CPU threads and GPU streams. Designed a per-schedule event pool to support concurrent execution of multiple scheduling strategies, and used cyclic signal chains with UBatchContext-based state management to avoid deadlocks.
- **Results**: Under the 8K input scenario, improved throughput from 0.22 QPS to 0.29 QPS, achieving a 31.1% increase; reduced average latency by 5.4% and P99 latency by 2.9%. The feature was deployed to TikTok's

Southeast Asia business during the 2025 Double 11 campaign, delivering over 20% online performance improvement.

Huawei 2012 Laboratories | Heterogeneous LLM Inference Framework 2025.05 – 2025.08

- **Challenge**: Large language model inference suffers from severe GPU memory bottlenecks. Existing approaches either sacrifice model accuracy, such as quantization, or introduce significant latency due to frequent CPU-GPU data transfers, making it difficult to satisfy low-latency and high-throughput serving requirements.
- **Technical Innovation**: Proposed xInfer, a collaborative inference architecture based on vLLM 0.5.5 that decomposes LLM inference workloads at the operator level. By dynamically identifying compute-intensive and memory-intensive operators, xInfer leverages idle CPU compute and memory resources to process part of the workload, enabling deep CPU-GPU collaboration and alleviating the bottlenecks of traditional offloading schemes.
- **Solution**:
  - Designed an operator-level dynamic scheduler that keeps compute-intensive operators on the GPU while partially offloading memory-bandwidth-sensitive attention operators to the CPU for parallel execution.
  - Built an asymmetric pipeline that alternates two mixed-computation batches across CPU and GPU, effectively hiding cross-device data transfer and computation latency while maximizing hardware utilization.
- **Results**: Evaluated the system on small-scale LLMs such as Llama3-8B, where the proposed asymmetric pipeline achieved a 1.18× throughput improvement.
- **Academic Output**: First-author manuscript submitted to ICPADS 2026.

Publications

- **Duo Xu**, Kunpeng Xu, Li Li. *Heterogeneous Collaborative Inference Architecture for Large Language Models*. Submitted to ICPADS 2026, first author.
- Kunpeng Xu, **Duo Xu**, Yuchen Wang, Li Li. *When Overlap Hurts: Closed-Form Scheduling Theory for Tensor-Parallel LLM Inference*. Submitted to EMNLP 2026, co-first author.

Technical Skills

- **LLM Inference Systems**: Familiar with vLLM, KV cache management, continuous batching, PagedAttention, prefix caching, tensor parallelism, and core mechanisms of LLM serving systems.
- **Performance Optimization**: Experienced with PyTorch Profiler, Perfetto, NCCL communication analysis, CUDA Runtime, CPU-GPU heterogeneous execution, and asynchronous scheduling.
- **Engineering Skills**: Proficient in Python, C++, and Shell, with experience in model serving, performance benchmarking, distributed debugging, and system-level experiment design.
- **Models and Frameworks**: Familiar with Llama, Hunyuan Image, PyTorch, Transformers, LMDeploy, vLLM-Omni, and related inference frameworks.

Additional Experience

- Part-time Campus Ambassador: Yanfu Investments, Tencent, Taotian Group, Meituan.